MASSACHUSETTS INSTITUTE OF TECHNOLOGY

IDSS
MIT INSTITUTE FOR DATA, SYSTEMS, AND SOCIETY

Great Learning
POWER AHEAD



WELCOME TO DATA SCIENCE & MACHINE LEARNING

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Some guidelines



Listen-only mode



Type your questions in Q&A box



Ask questions which are in the interest of the larger audience

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Agenda

- Program Manager & Panelist introduction
- Learning philosophy and Collaboration b/w IDSS & GL
- Program details
- Q&A
- Intro to the world of data science

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DATA SCIENCE &
MACHINE LEARNING

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Program Manager



Vartika Misra
August 2023 - B

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DATA SCIENCE &
MACHINE LEARNING

DSML August 23 Cohort

340+ Learners

40+ Countries

25+ Industries



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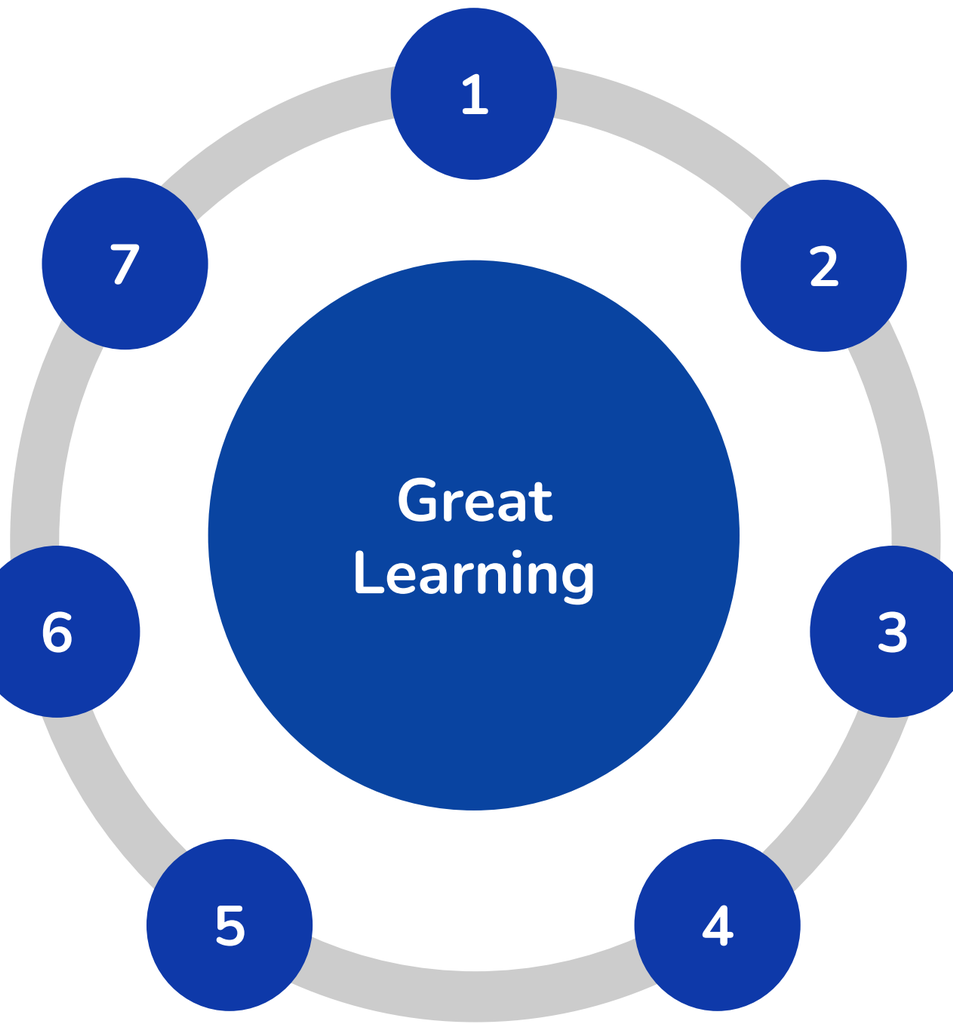
Collaboration: MIT & Great Learning

- Academic Collaborator
  - Curriculum & content design
  - Recorded Video lectures
  - Case studies / Projects
  - Certificate
- Delivery Collaborator
  - Mentored learning
  - Learning management system
  - Program Manager
  - Academic support

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About Great Learning

Established in
2013



Continuous **Innovation** & **Improvements**

20+ Mn hours of learning

Olympus, our in-house proprietary LMS

25k+ Working professionals

2000+ gurus, Mentors & Academicians

Established track record of **program completion**

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The Learning Philosophy

- Have a clear learning path that is well-structured, as well as comprehensive
- Access to content from credible academic faculty
- Learning by doing - practice case studies with actual problem-solving
- Have access to a mentor who can clarify doubts & provide industry perspective
- Be tested frequently & get personalized feedback to be aware of the progress
- Have a support available during tough times
- Have access to peers to network

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Approach to learning

Traditional Classroom

- Instructor prepares material to be delivered in class
- Learners listens to lecture in-class, and other guided instructions and takes notes
- Homework and assignments to demonstrate learning

Flipped Classroom

- Instructor records and shares lecture outside of class
- Recorded Lectures and content to be watched before the session
- **In-class l**earning includes applied learning and **higher order thinking tasks**
- Support through peers and Mentor

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DSML August 2023 Delivery Schedule

	Course	Topic	Content Release Date	Assessment Deadline	Session Starting (Sat)	Mentored Learning Session
Preparatory sessions	Into the world of Data Science	Pre work - Introduction to Python	Available on enrollment	-	15-Jul-2023	No
		Intro to the world of Data Science	Available on enrollment	-	22-Jul-2023	No
		Pre-work - Introduction to Python	Available on enrollment	-	29-Jul-2023	No
		The Must-know Mathematics and Statistics Behind DS	Available on enrollment	-	5-Aug-2023	No
		Program Orientation + Intro to the world of Data Science	Available on enrollment	-	12-Aug-2023	No
		Pre-work - Introduction to Python	Available on enrollment	-	13-Aug-2023	No
		Data Science Application Case Study & Statistics	Available on enrollment	-	19-Aug-2023	No
	Course	**Topic**	**Content Release Date**	**Assessment Deadline**	**Session Starting (Sat)**	**Mentored Learning Session**
Week -1	Foundations of Data Science	Python for Data Science	Available on enrollment	27-Aug-2023	26-Aug-2023	Yes
Week -2		*Learning Break*			2-Sep-2023	
Week -3	Foundations of Data Science	Statistics for Data Science	24-Aug-2023	10-Sep-2023	9-Sep-2023	Yes
		Project 1	24-Aug-2023	10-Sep-2023		
Week -4	Making Sense of Unstructured Data	Introduction, Clustering, Spectral Clustering, Components, and Embeddings	7-Sep-2023	17-Sep-2023	16-Sep-2023	Yes
Week -5		*Learning Break w/ Hands-on Masterclass 1*			23-Sep-2023	
Week -6	Regression and Prediction	Classical Linear and Nonlinear Regression and Extensions, Modern Regression with High-Dimensional Data, The Use of Modern Regression for Causal Inference	14-Sep-2023	1-Oct-2023	30-Sep-2023	Yes
Week -7		*Learning Break w/ Hands-on Masterclass 2*			7-October-2023	
Week -8	Classification and Hypothesis Testing	Hypothesis Testing and Classification	28-Sep-2023	15-Oct-2023	14-Oct-2023	Yes
		Project 2	28-Sep-2023	15-Oct-2023		
Week -9		*Learning Break w/ Hands-on Masterclass 3*			21-October-2023	
Week -11	Deep Learning	Deep Learning	12-Oct-2023	29-Oct-2023	28-Oct-2023	Yes
Week -12	Recommendation Systems	Recommendations and Ranking, Collaborative Filtering, Personalized Recommendations	26-Oct-2023	5-Nov-2023	4-Nov-2023	Yes
		Project 3	26-Oct-2023	5-Nov-2023		
Week -13	Networking and Graphical Models	Introduction, Networks, Graphical Models	2-Nov-2023	-	11-Nov-2023	Optional Masterclass
Self-paced	*Self-paced Modules*					
	Predictive Analytics	Predictive Modeling for Temporal Data, Feature Engineering	9-Nov-2023		18-Nov-2023	Optional Masterclass
	ChatGPT and Generative AI	Transformers, Large Language Models	9-Nov-2023		-	-
		Hackathon	17-Nov-2023	19-Nov-2023		
		Program Completion Ceremony			25-Nov-2023	

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Weekly Operating Rhythm



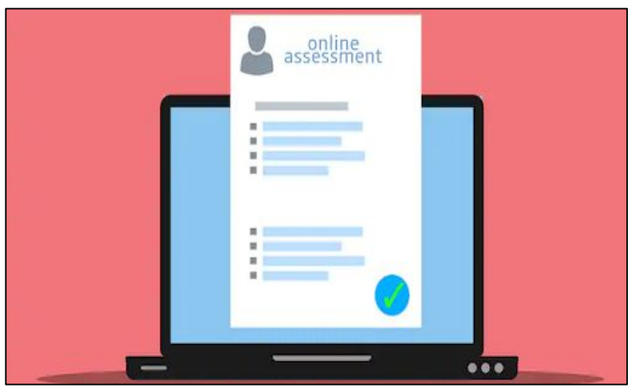
Weekdays (Monday - Friday)

- Pre-reads
- 2-3 hours Recorded faculty video lectures
- Practice exercises



Weekends (Saturday - Sunday)

- Mentored learning session
- Hands-On Master class during learning break
- Optional Masterclass for the self-paced module



Assessment

- Weekly course assessments
- Projects with weekly deadlines in 3 courses

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Assessments are weekly

In order to be eligible for the certificate, you will have to complete all courses with minimum of 60% marks in each course.

Weekly Course Assessment

- 7 deadline driven Course assessment
- 1 assessment per week

Projects

- 3 deadline driven projects
- 1 project each in the following course: Foundations of Data Science, Classification and Hypothesis Testing & Recommendation Systems

***Assessments in every course carry 90% weightage**

***Attendance of the mentored learning sessions carries 10% weightage**

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Mentored Learning Session Structure

8-10 mins — Mentors gauge the **level of understanding** of learners & identify areas to focus on

45 - 55 mins — Clarification **of doubts** on concepts/learning from the course of the week

45 - 55 mins — **Case Study** discussion during the session

15 - 25 mins — **Industry Perspective** through additional Q&A

2 - 5 mins — **Summarize** the Session

Q&A & Industry Perspective

Can run throughout the session as it is meant to be an interactive one

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Mentored Learning Groups

- ~18-22 learners in a group
- Factors:
    - Geography and your time zone
    - Profile - Programming experience/Work experience
    - Academic Background
    - Mentor availability
- Either Saturday or Sunday

Timings will be communicated in the coming week

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Additional Webinar Sessions

Hands-On Masterclass:

- Focus on implementation of the conceptual knowledge gained in the previous week towards solving a problem based on real-world data.
- Focus on the practical application through case studies for the topics of the previous week, these sessions further clarify the content from a programming perspective.
- 3 Hands-on masterclasses on the following topics: Making Sense of Unstructured Data, Regression & Prediction, and Classification & Hypothesis Testing
- 2-hour session in a webinar format, dedicated to exploring the application of these concepts to a hands-on business case study
- Conducted by an industry expert for the cohort in a Masterclass format
- Attendance doesn't carry weightage in the marks for the course.

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Learning Support Ecosystem



Olympus

- You can raise support request through Olympus
- All queries are responded to within 24-48 hours
- All program announcements show as notifications

WhatsApp

- Peer learning through groups & discussion forums
- Ad-hoc communications from Program Office

Program Manager

- Direct call or text or calendar invite
- Available during office hours
- Single POC for all support from the ecosystem

Program Office Email

- Prefered mode for candidature change actions
- Onboarding & post program communications
  (office.dsml.mit@mygreatlearning.com)

Faculty

Mentors

Peer Group

Program Manager

Academic Assistance

GL Community

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How to learn more effectively?

Commit to the journey

- 8-12 hours per week
- Follow the weekly operating rhythm
- Attend the mentored learning sessions
- Respect Assessments

Make the most of the content

- Take detailed notes of videos
- Raise support queries on Olympus wherever you need help
- Go through all the additional reading materials shared

Prepare well for all sessions

- Go through the weekly content before the sessions

Practice, Discuss, Repeat

- Go through the case studies & practice them
- Discuss with peers & mentor
- Read, explore & concretize

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Give us a lot of feedback

- Be ***Descriptive*** – Take the time to detail your feedback
- Be ***Constructive*** – How can your learning be improved?
- Be ***Specific*** – Use instances, examples, etc.
- Be ***Realistic*** – We are balancing the whole class' needs

CALL OUT >> WE ASK FOR A LOT OF FEEDBACK

These are discussed in the leadership meetings

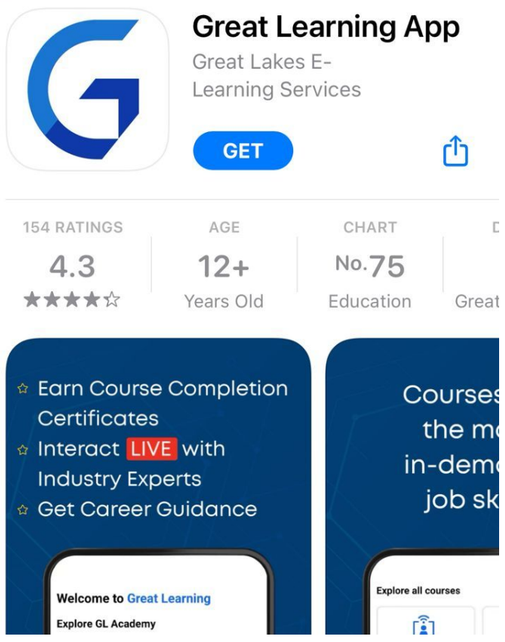
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Tools & Technology

olympus.mygreatlearning.com



Great Learning App



Online Sessions



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Additional Features



Resume and LinkedIn Profile Review

- Feedback from experts to make your resume and LinkedIn profiles stand out from the crowd.
- Tips on customizing your resume for the job profile you are applying to.

GL Connect

- GL Connect is where you can message and connect with your cohort's learners.
- It will be accessible post completion of the program

GL Community

- GL Community is our peer connectivity Platform with over **10,000 members**
- It will help you to start a Discussion or Write a Blog
- **Practice** your skills by answering Questions on the platform
- **Engage**, learn and connect with your peers and experts

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Next Steps

Login to Olympus - olympus.mygreatlearning.com

- Complete your profile on Olympus
  (Name, linkedIn profile, contact details)
- Update your time zone on Olympus

Go through the following courses

- Pre-work
- Foundations of Data Science
- Pre-work : Introduction to Python session scheduled on Sunday i.e. 13th August.

- **Join the first mentored learning session on the weekend of August 26/27th *(Link will be available under UPCOMING/ONGOING Activities on dashboard)***

- **Submit the weekly quiz before the deadline & in case of any questions, kindly raise a support request on Olympus through the 'Need Assistance' option**

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Thank You

We'd love to hear your feedback!

Please share your feedback for the Orientation session

Wish you all the very best!

Please feel free to raise a support request through Olympus

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